

Grant identification number (TFR09-01213)

Sri Lanka Wildlife Conservation Society

ele/D Project

Summary Report

An Automated Elephant Recognition System

Introduction

Our goal

To provide a cost effective and non-invasive method to gather ecological information of elephants by developing technology that is user friendly and open to citizen science.

For the conservation and management of elephants, having an efficient system to monitor populations, ranging, habitat use, and distribution is important. However most of the existing methods to obtain this information are expensive, not open to easy collaboration and time intensive. In addition some of these methods are invasive and can adversely affect the natural life style of elephants. Our goal is to come up with a non-invasive and efficient mechanism to, identify and track elephants for the purpose of conservation.

Objective of the project

To develop '*ele/D*' as a non-invasive computer vision based mechanism to identify and track individual elephants. People from all around the world can contribute to this and get involved in it via a web portal, which will be integrated with Google Earth.

ele/D Project Progress

ele/D system (in brief)

The *ele/D* system carries out individual elephant recognition utilizing user input images and direct user inputs, to develop a constantly updated elephant image and spatial and temporal information monitoring automated system and database. As the system evolves, the logged data will help to extract demographic and behavioural information on elephants which are crucial for

elephant habitat protection and to mitigate human elephant conflicts, and it will be merged with *elePortal*, which will form a user community of people interested in elephant research and conservation. The method of automated elephant identification is based on the texture based technique of eigenfaces for recognition.

Methodology

eleID recognizes elephants by combining two pieces of information, automatic elephant recognition, and direct user inputs. When an image of an elephant is uploaded into the system, an image based algorithm searches for and matches the image with an existing annotated elephant database. If it is a previously identified individual, then the system will match the image with information available for that particular elephant. Along with this closest match the system will allow the user to verify whether the best match is correct or the next possible is the best candidate. If the elephant's identity is confirmed then the database gets updated with the last observed location and the date. If the elephant is previously unidentified, then the system issues a new identity. Utilizing the Google Earth API, researchers and the general public from all around the world will be able to contribute to this process by uploading frontal images of elephants that they have observed. Along with the images through the *elePortal* users can give additional information about the elephant, such as relative tail length, tail tuft information, and details of primary and secondary ear folds. This additional information of an elephant is used with an evidence combining mechanism to recognize individual elephants. *elePortal* will also include secondary data such as socio-economic and human elephant conflict information that is tagged to geographic locations so that users can analyse trends and use the system for research, conservation and development needs.

We have adopted a vision based mechanism to identify elephants from their frontal face. The vision algorithm is based on Principal Component Analysis (PCA) more specifically known as Eigenface based processing. Getting the direct correlation between each and every elephant face image is a cumbersome exercise which is not efficient. Due to that reason here we do a dimensionality reduction by applying PCA to elephant images. This gives us only the essential (orthogonal) dimensions which are required to distinguish one elephant image from another. These orthogonal vectors or the principal dimensions are known as Eigenvectors in the vision community. For each elephant face image we can get the weight to it from each and every eigenvector. Once we get the weights from each eigenvector for a particular image, we can use that weight vector (which is just an array of floating point values) to find the relationship between elephant images. In other words here we try to match the nearest neighbor to a particular elephant face image. If the distance to the nearest neighbor is very low, then there is a higher possibility that those neighbors correspond to the same elephant. After presenting the result to the user we can get a verification in regard to the accuracy of the result. But if the nearest neighbor is not so close to a particular elephant face image which need to be identified we could say that there is a higher possibility of the elephant shown in the image being an unidentified elephant. This has to be verified by the user.

Experiments and Results

Our eigenface based identification mechanism for elephant identification was tested on frontal face images obtained from a database of African elephants (*Loxodonta africana*). The accuracy of identification using only the frontal images was around seventy percent (error percentage of the accuracy rate $\pm 3\%$).

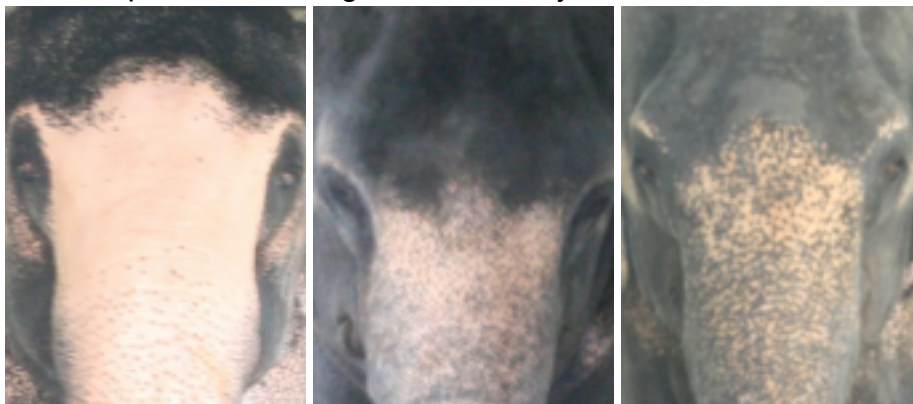
Evaluation process

To evaluate the process we used a properly annotated database (Bulls Prime) which contains categorized images of 192 African elephants (with couple of images per each elephant). We used some images to train the system and separate image sets to test the identification process. By giving images of known elephants we measured the percentage of accuracy in every major milestone in the implementation of the core project.

Related Issues

In the initial, purely individual image based elephant identification mechanism, the reliability of the identification process relied mainly on the quality and exact frontal face images of elephants. The above test was carried using all the images without any modification and once we processed them the accuracy can be further improved. The frontal image that is taken should represent an average view of the elephant face without having any strange poses, which is hard to fulfill in real situations. Another crude assumption is illumination invariance, because particularly in the field, capturing an image of a wild elephant under uniform illumination is impractical.

The following is a sample of the frontal elephant images that we present to the recognition algorithm. As mentioned earlier these images have been taken without any modification. Intuitively we can say that, if we do some preprocessing we can improve the recognition accuracy.





When it comes to users from around the world uploading images and other data related to elephants via *elePortal*, we cannot always assure the reliability of the data even though the image quality can be handled up to some extent by our image based identifier. In order to make the system robust these issues have to be handled.

Significant challenges of *eleID* project and how we dealt with them

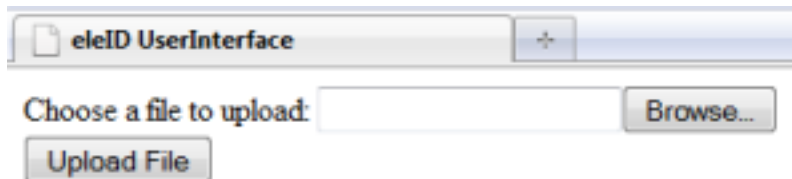
- Changing the identification strategy: Even though we initially thought of using nodal point analysis for identification purpose (like the relative ratios between unique points) due to inefficiencies in algorithms we thought of changing the direction to a texture based identification mechanism.
- Not solely depending on image identification: Our initial plan was to develop a robust image based identification system which has the accuracy of more than 95%. But with the results obtained in the intermediate stages our strategies have changed in order to cater the aid of other supportive information such as, observed location, special marks unique to the elephant, etc. In the future we hope to focus on a proper decision making algorithm to assist the system in giving the right identification considering all the evidences.
- Trusting the images and information given by the users: When users upload the images and give observed location/data information, there is a risk involved in accepting the data as 100% accurate, which might not be the situation in reality. To solve this issue what we have thought of was categorizing the user community in to several categories (e.g. administrators, scientists, researchers, volunteers, general public etc) and make a priority scheme for the sources of information.

Current Direction

Currently we are focusing on two parallel paths. In one path we are working on improving the accuracy of the identifier. This includes some specific preprocessing stages prior to identification. In the second parallel path our objective is to come up with a working online system. So by the time we come up with an enhanced identifier we will be ready to run it online in a very short time.

The following shows the three basic stages of elephant identification process that we have developed so far.

Stage 1 – Users from all around the world upload elephant images. This interface will be integrated with *elePortal* and in that situation additional information will also be obtained from users like observed location and date.



(once 'Browsed' the following window will appear)

Stage 2 – User has to click on the eye positions of the elephant in the uploaded image

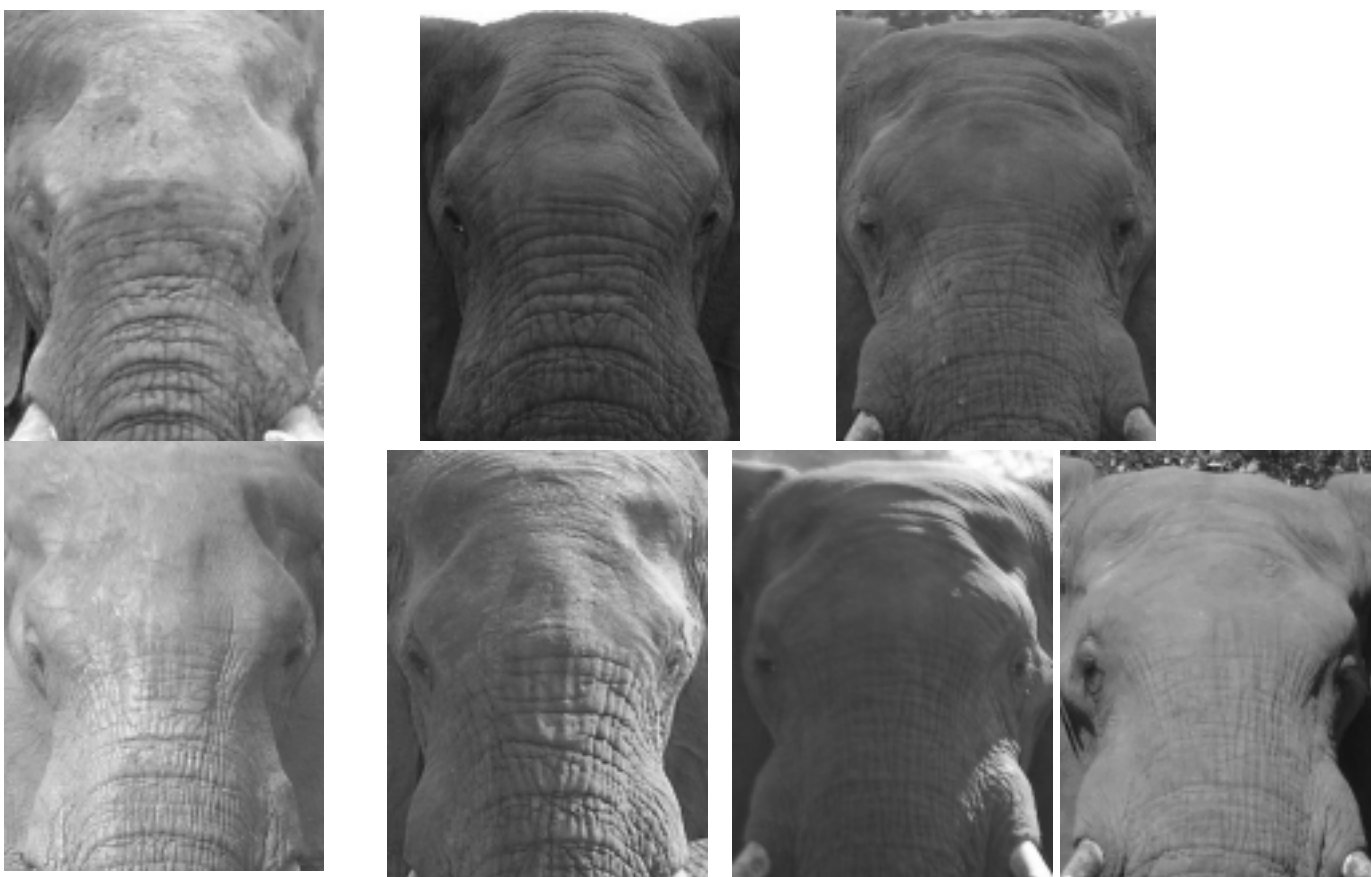
Right side eye

Left side eye

(Note: The left side is as user sees the image and not really elephant's left eye. The same rule applies for right side.)

Stage 3 – The results will be displayed (currently we are using a Google Map to show the results).

As a solution to handle scaling and rotation issues we are asking the user to select the eye positions of the elephant in the image. This allows the system to identify the elephant in the user given image, even in situations where there are multiple elephants in the same image. Also if we try to automatically detect the elephant faces in images, it will give us the additional risk of false negatives (not detecting the elephant face which is actually in the image) and false positives (false detection of elephant faces when there are actually no elephants in the image), which in turn affects the overall accuracy of the system. The following images show some of the results that were obtained from an intermediate stage (these images are in gray-scale). At present we are working on adding more preprocessing stages so that it will increase the accuracy.



Assistance Needed from Google.

For this project, we would appreciate very much the support from Google Team to make the system operate online. So far we have completed a core system, which has been used to test a basic online system. The figure shown below gives the overall idea of the system.

Assistance required from Google Team in regard to Second Module

Future Work

In addition to using image based evidence to identify individual elephants, additional user input data on relative tail length, nature of primary, secondary ear folds along with the shape of ear bottom, presence of tusks, tail tuft, special markings such as rips, holes and notches can be incorporated for identification. For this to be done, a proper evidence combining mechanism has to be used. When using such evidence combining mechanism, human errors can also be minimised by analysing the user capabilities and reliability of the given data.

Due to its accessibility to a large number of users from around the globe who can upload images to it from anywhere, the *e/eID* system is open to citizen science and eliminates one of the major deficiencies in manual image based elephant identification. In manual methods, the researchers or the team involved in the identification task have to recognize from memory each and every individual elephant identified. Such a method is time consuming and hard to use for collaborative work. As a future direction, this system

can be easily adapted to help gather habitat use and population information of other endangered species for their conservation and management.

Once we have a working online system it would be easy for any person to contribute to this project. In the future enhanced identification algorithms can be developed and replace the existing core to increase the accuracy of the identifier.

What we have learned

- Importance of community involvement: From the projects that we have carried so far, including especially the *eleID* project, what we have realized is the power of the involvement of the 'general public'. This simply means that there is a vast majority of people around the world who would really like to contribute to and commit themselves to research that supports elephant conservation. Unfortunately the infrastructure mainly in the form of "user friendly" technology is not available or has not been developed yet. Through our online based *elePortal* we hope to open our *eleID* project to all the people around the world, providing an essential bridge that will link community participation with conservation activities. Developing such a link will add tremendous stimulation to the growth of the *eleID* project increasing its potential to generate support for Asian elephant conservation globally.
- Focusing on machine learning approach in future implementations: After series of different experiments some schools have claimed that, the answer to certain solutions can be easily obtained from letting the machine learn about it rather than giving rules for everything (which leads to exceptions all the time). Our technical team in the *eleID* project has taken this fact into serious consideration and future research and development efforts are going to be in that direction.
- The concept of 'One-Stop-Shop': Our future vision is to make *eleID* and *elePortal* "one stop-shop" by providing an easy and convenient to use online resource base to access information related to everything pertaining to elephants.

Publications and Collaborations

- April 2010 - Summary Report (this document)
 - Feb 2010 - Initial collaboration discussions with International Elephant Foundation, Elephants for Africa, Save the Elephants, Elephants without Borders, GRASP - UNEP/UNESCO Great Ape Survival Partnership and Bristol University (Penguin ID project).
- Jan 2010 - Presented at International Elephant Conservation and Research Symposium in South Africa.
<https://docs.google.com/presentation/edit?id=0AfqWjQdmJ9U5ZGNidGs1Y3JfODVmZ3RxMjhnZW&hl=en>
- Dec 2009 - Video by Greenermedia -
<http://www.kickstarter.com/projects/greenermedia/asian-elephant-conservation-documentary?pos=1>
- Sept 2009 - Poster displayed at SLWCS 10th Anniversary Celebrations -

- www.slwcs.org/downloads/posters/leID-Poster-Sept2009.jpg
and www.slwcs.org/downloads/posters/elePortal-Poster-Sept2009.jpg
- Aug 2009 - Press release https://docs.google.com/View?docID=0AfqWjQdmJ9U5ZGNidGs1Y3JfNmM0bnJkdGdo&revision=_latest
 - July 2009 - Presentation at IUCN, AsESG, Human Elephant Conflict Task Force Workshop and Society for Conservation Biology Annual sessions in Beijing, China.
 - Jan 2009 - SLWCS 2007-2008 Annual Report

Milestones

April 2010 - A Vision based elephant identification method developed and tested. Aug

2009 - A Vision based literature survey has been done in regard to elephant

recognition. June 2009 - Mr. Ranga Dabbare hired to work on the *eleID* project full

time for his MSc thesis.

June 2009 – Ms. Ayesha Matarage hired part time to work on the *elePortal* component of the project (3 months).

May 2009 - Tides Foundation and Google Earth finalize grant to begin work on the

system. Nov 2008 - Initial discussion with University of Moratuwa to develop

system.

Sept 2006 - Dec 2007 - volunteer team initiates work using facial recognition systems (VB system developed).

July 2006 - SLWCS Board and Volunteer biologists start to develop basis for

eleID. [1] HEC stands for Human Elephant Conflict